LESSON 2.5

Describe Data by Range and Interquartile Range

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

WHY THIS MATTERS The league analyst is comparing two sixth-grade classes that took the same math test. Both classes have the SAME median score, 81 — so the medians alone cannot tell them apart. Your job is to measure how spread out each class's scores are, using the range and the interquartile range, and decide which class did better overall.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Range and Interquartile Range *Spanish: Rango y rango intercuartílico*	Two numbers that describe spread: the range covers all the data, and the interquartile range covers only the middle half.	Range 26 covers 68 to 94; IQR 12.5 covers only 75.5 to 88
Range *Spanish: Rango*	The distance from the least value to the greatest value, found by subtracting: greatest − least.	greatest − least
Quartile *Spanish: Cuartil*	One of the three values that cut an ordered data set into four equal-sized groups.	Q1, median, Q3
Interquartile range *Spanish: Rango intercuartílico*	The distance across the middle half of the data, found by subtracting the first quartile from the third: Q3 − Q1.	IQR = Q3 − Q1
Variability *Spanish: Variabilidad*	How much the values in a data set differ from one another — whether they sit close together or far apart.	Close together or far apart
Measure of variation *Spanish: Medida de variación*	A single number that describes the spread of a whole data set. The range and the IQR are both measures of variation.	Range and IQR are both

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Hikes (km): 5, 12, 8, 20, 15, 10, 18.

1. Order: 5, 8, 10, 12, 15, 18, 20.
2. Range = 20 − 5 = 15 km.
3. Q1 = 8, Q3 = 18, so IQR = 18 − 8 = 10 km.

Answer: Range 15 km; IQR 10 km

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Class A's box plot reads: least 62, Q1 74, median 81, Q3 87, greatest 90. Find the range first.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The greatest value is _____ and the least value is _____ .
- The range is _____ , which describes _____ .
- Quartile 1 is _____ and quartile 3 is _____ , so the IQR is _____ .
- The IQR describes only _____ of the data.

Word bank range | interquartile range | quartile | spread | variability | greatest | least | subtract | middle half | extreme | order | consistent

WATCH OUT!

A common mistake

A common mistake with Range and Interquartile Range is adding the two quartiles instead of subtracting them

LESSON 2.5

Describe Data by Range and Interquartile Range

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Two classes both have a median score of 81. Class A's IQR is 13 and Class B's IQR is 8. Which statement is true?

(A) Class B has a higher median

(B) Class A has more students

(C) The middle half of Class B's scores is less spread out

(D) Class B's highest score is greater

HINT A smaller IQR means a narrower box.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each fact under the measure of variation it describes.

Groups: **Range** **Interquartile range**

_____ Greatest value − least value

_____ Q3 − Q1

_____ The width of the box on a box plot

_____ The distance from whisker tip to whisker tip

_____ Changes a lot when one value is unusually high

_____ Describes the middle 50% of the data

HINT Ask which values each measure subtracts.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A data set runs from a least value of 12 to a greatest value of 30. What is the range?

(A) 12

(B) 18

(C) 21

(D) 30

HINT Range asks how far it is from one end of the data to the other.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

4 **Circle the letter of the best answer. Show how you know.**

On a box plot, Q1 = 20 and Q3 = 36. What is the interquartile range?

(A) 16
(B) 28
(C) 36
(D) 56

HINT The IQR is the width of the box on a box plot.

Put the data in order _____

WORK

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

5 **Circle the letter of the best answer. Show how you know.**

Which measure of spread uses ONLY the middle half of the data?

(A) The interquartile range
(B) The range
(C) The median
(D) The mode

HINT One is measured across the box on a box plot; the other whisker to whisker.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** Data: 40, 42, 45, 50, 55, 58, 60. Q1 = 42, median = 50, Q3 = 58.
2. **Step 1** Range = 60 − 40 = 20
3. **Step 2** IQR = 58 + 42 = 100
4. **Step 3** The middle half of the data spans 100 units.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Compare the IQR in Step 2 to the range in Step 1.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

SENTENCE STARTER I would use the ___ because it only uses the ___, so one unusual value cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.5

Describe Data by Range and Interquartile Range

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each value from the history test data (68, 70, 73, 74, 75, 76, 77, 80, 82, 83, 84, 85, 86, 86, 88, 88, 90, 90, 93, 94) with what it is.

_____ 26

_____ 12.5

_____ 83.5

_____ 75.5

_____ 88

(A) Range

(B) Interquartile range

(C) Median

(D) First quartile

(E) Third quartile

2 **Circle the letter of the best answer. Show how you know.**

Team A's salaries have a range of $14M and an IQR of $7M. Team B's have a range of $7M and an IQR of $2.5M. Both teams have the same median salary. What does this tell a player?

(A) Team B pays more overall

(B) Team B's salaries cluster much more tightly around the median

(C) Team A has more players

(D) Team B's highest salary is greater

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

A value far above the rest is added to a data set. Sort what happens.

Groups: **The range grows a lot** | **The IQR barely changes**

_____ Because it uses the greatest value

_____ Because it uses only Q1 and Q3

_____ It measures the whole span of the data

_____ It measures only the middle half

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A box plot shows: least 68, Q1 75.5, median 83.5, Q3 88, greatest 94. What are the range and the IQR?

(A) Range 12.5, IQR 26

(B) Range 20, IQR 12.5

(C) Range 26, IQR 12.5

(D) Range 26, IQR 83.5

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Class A's test scores are shown on a dot plot. Most dots stack between 80 and 90, and a few dots trail off to the left at 52, 58, and 61. Which word describes the SHAPE of this distribution?

(A) Skewed left — the long tail points toward the lower scores

(B) Skewed right — the long tail points toward the higher scores

(C) Symmetric — the left and right sides match

(D) Uniform — every score appears about equally often

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** Data: 15, 18, 20, 22, 25, 60.
2. **Step 1** The range is 60 − 15 = 45.
3. **Step 2** The range of 45 shows that a typical value differs from the others by about 45.
4. **Step 3** So this data set has high variability everywhere.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

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LESSON 2.5

Describe Data by Range and Interquartile Range

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

A coach wants ONE number for each question. Sort each question under the measure that answers it.

Groups: Range | Interquartile range

_____ How far apart are our best and worst games?

_____ How consistent is a typical night for this team?

_____ How much did the middle half of our scores vary?

_____ What is the full span from lowest to highest?

_____ Which measure survives one record-breaking game?

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** Class A: least 62, Q1 74, median 81, Q3 87, greatest 90.
2. **Setup** Class B: least 70, Q1 76, median 81, Q3 84, greatest 90.
3. **Step 1** Both classes have a median of 81.
4. **Step 2** Both classes reached the same greatest score, 90.
5. **Step 3** So the two classes performed the same overall.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

Two students both scored 80 on a make-up test — one joins Class A (range 28, IQR 13) and one joins Class B (range 20, IQR 8). Both classes have a median of 81. Predict how each class's IQR changes, and explain your reasoning.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Both numbers describe spread. Why might a coach trust the IQR more than the range when one player had a record-breaking night?

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